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CRISPR Informatics Course

28-Jul-2025 to 01-Aug 2025

CRISPR Informatics Course

Determining Indels generated by CRISPR-Cas9

and

Validation of knock-out / knock-ins

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30-Jul-2025

What to expect in this session

The session will cover the following :

- Basics of gene editing using CRISPR-Cas9
- Brief overview on methods to determine Cas9 induced breaks
- Focus on using freeware tool – TIDE to determine editing efficiencies and outcomes
  - Practical demonstration
- Brief overview on using Cas9 to facilitate- gene knock-in / tagging
  - Practical demonstration

After this session you should be able to :

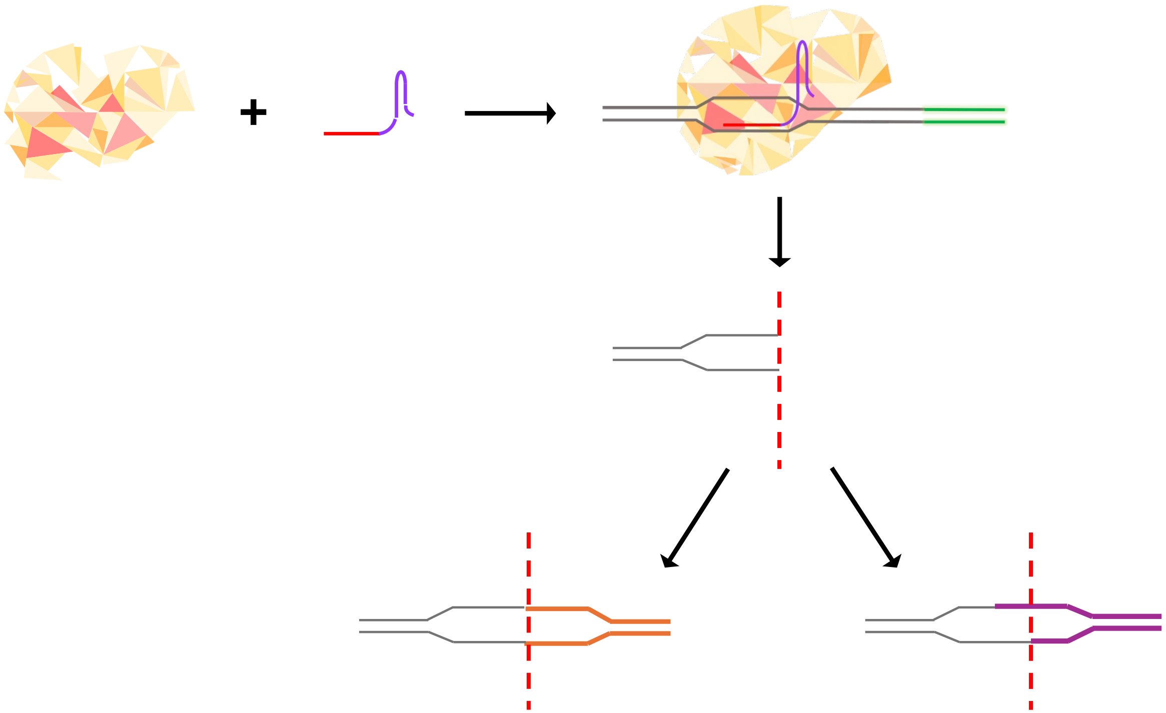
- Use TIDE to determine gRNA efficiency in your cell lines / samples
- Understand information contained on sanger sequencing trace files
- Determine knock-out/ knock-in/ SNV variations occurring with genomic region of interest

CRISPR-Cas9 editing

Clustered regularly interspaced short palindromic repeats (CRISPR) and CRISPR associated proteins (Cas) derived from microbial adaptive immune system to protect against viruses and plasmids.

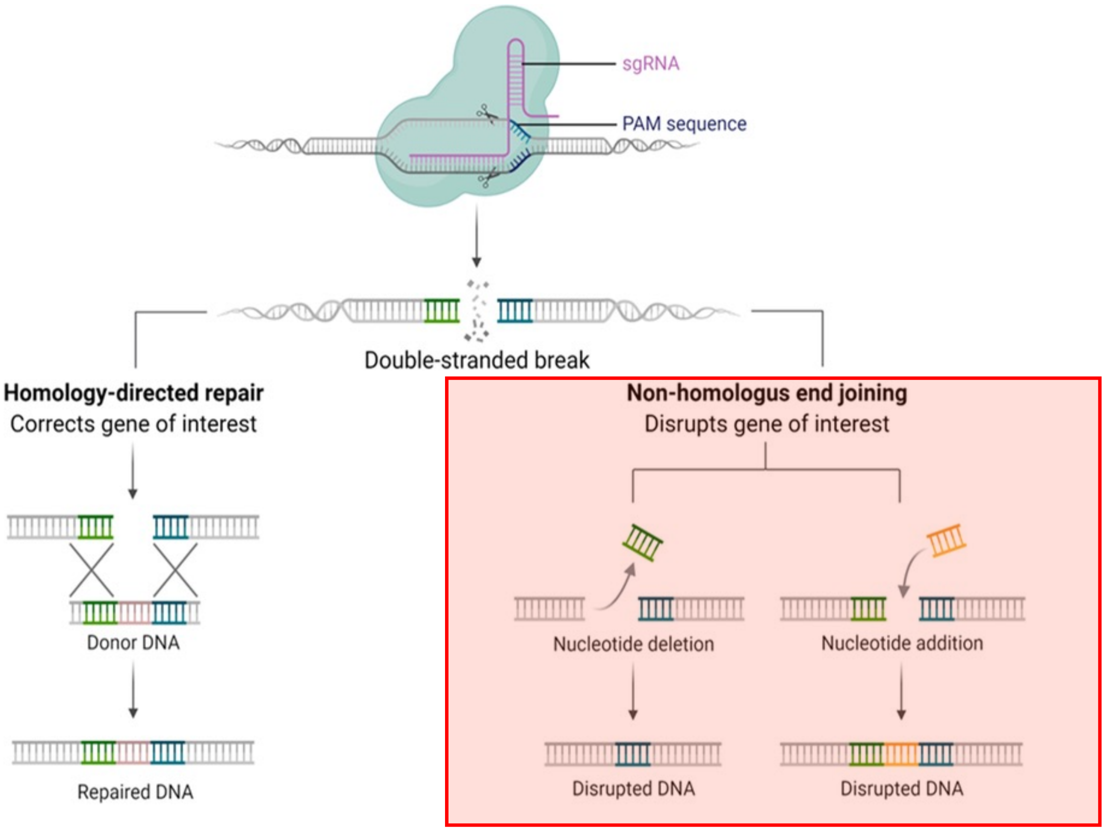
Cas9 is the endonuclease that can generate DSB when guided by the target sequence (gRNA)

gRNA / sgRNA is a short synthetic RNA composed of a scaffold sequence necessary for Cas-binding and a user-defined ~20 nucleotide spacer that defines the genomic target to be modified.



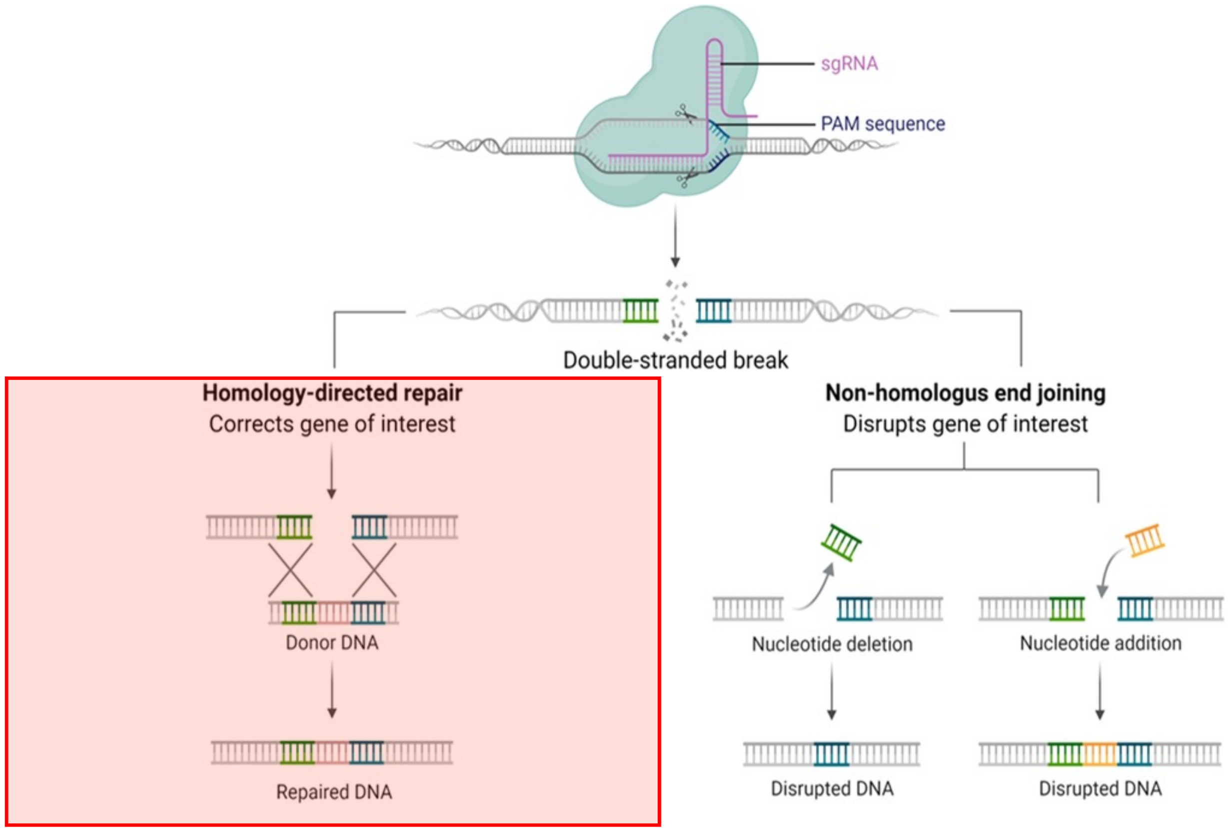
CRISPR-Cas9 editing – Generating Gene KOs

CRISPR-Cas9 system can be used to generate knock-out cell lines. An insertion or deletion induced by a single guide RNA is often used to generate knock-out cells, however, some cells express the target gene by skipping the disrupted exon, or by using a splicing variant, thus losing the target exon.



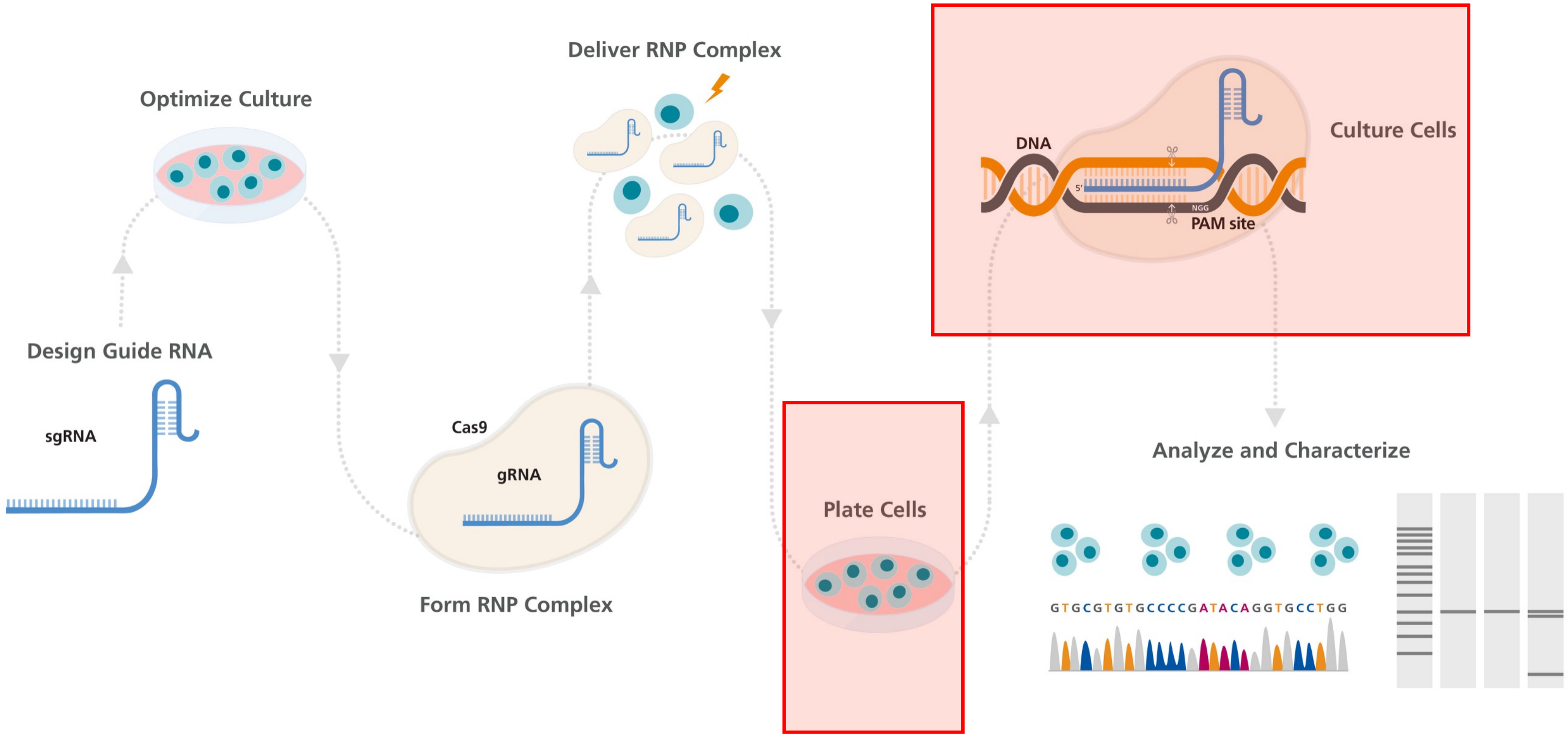
CRISPR-Cas9 editing – Generating novel lines

CRISPR/Cas9 gene knock-in or gene replacement system is designed to facilitate the insertion, removal, or replacement of a specific gene(s) within a given genome. The site-specific insertion of protein tags, modified promoters, and other regulatory sequences are also possible.



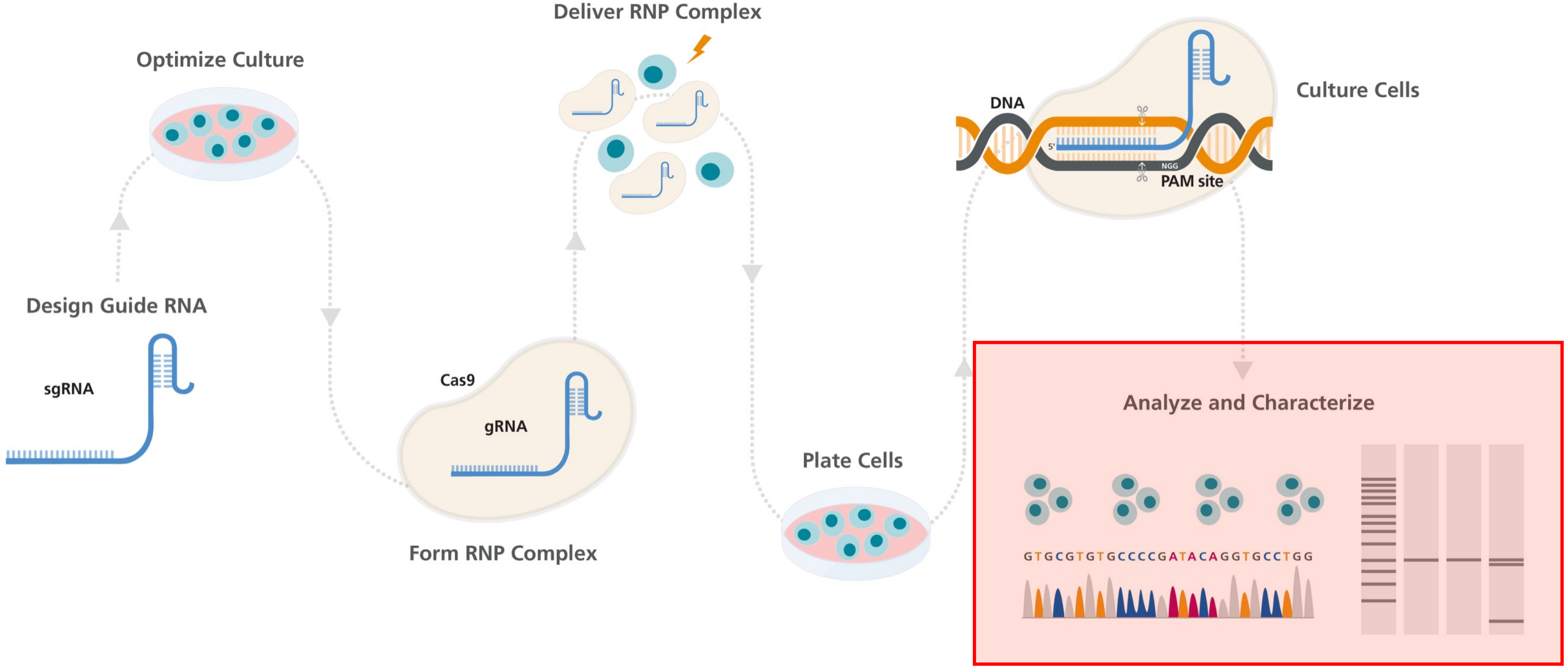
CRISPR-Cas9 edits - how to validate them

The simplest but often the least specific method of identifying successful CRISPR genome editing is to observe phenotypes of edited cells. In some experiments, the expected phenotype from the gene editing process is known.



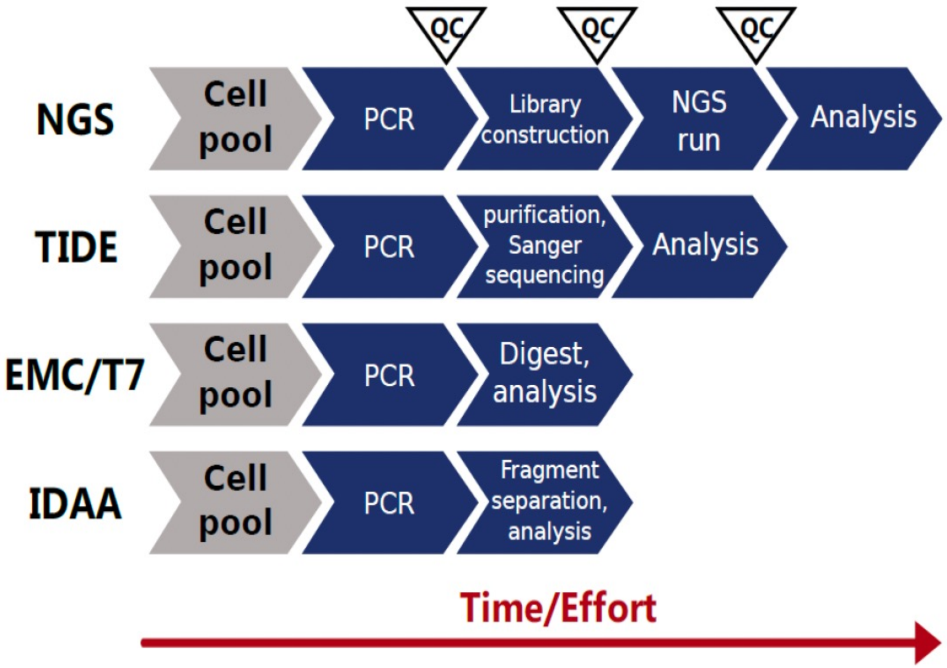
CRISPR-Cas9 edits - how to validate them

A more in depth view of the genomic edit that translates to the phenotype can be determined by investigating the genetic contact in and around the edited sites.



CRISPR-Cas9 edits – what are the methods are available and what do they provide

To determine the edits and its subsequent effects a range of techniques are available

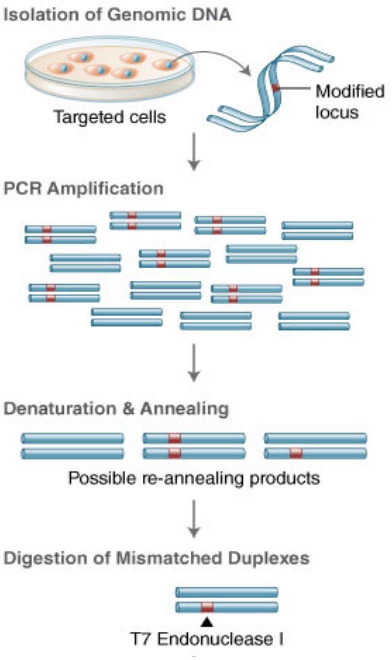


	Sensitivity	Resolution	Scalability	Cost
NGS	**high**	**high**	low	high
TIDE	medium	medium	medium	medium
EMC/T7	low	low	**high**	**low**
IDAA	**high**	medium	**high**	**low**

Understanding the outcomes of genome editing

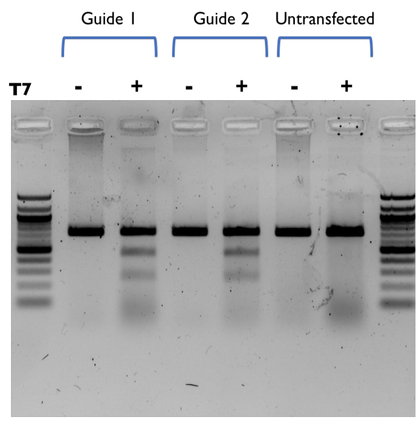
Surveyor nuclease assay

- **Surveyor** nuclease **assay** is an enzyme mismatch cleavage **assay**
- Can detect single base mismatches or small insertions or deletions (indels).
- **Surveyor** nuclease is part of a family of mismatch-specific endonucleases that were discovered in celery (CEL nucleases).



T7 endonuclease 1

- T7 Endonuclease I recognizes and cleaves non-perfectly matched DNA.
- PCR products are annealed and digested with T7 Endonuclease I.
- Fragments are analyzed to determine the efficiency of genome targeting.



HEK293 cells

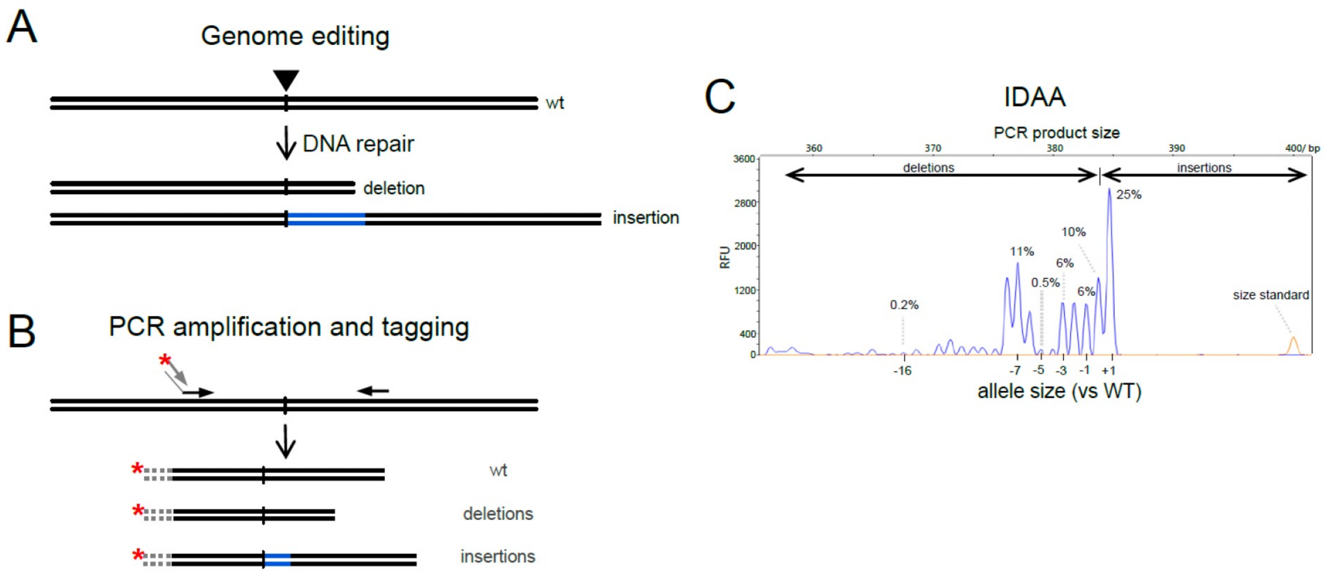
Understanding the outcomes of genome editing

IDAA (Indel detection by Amplicon Analysis)

- An alternative non-sequencing based proprietary method that enables sensitive and quantitative detection of InDels.

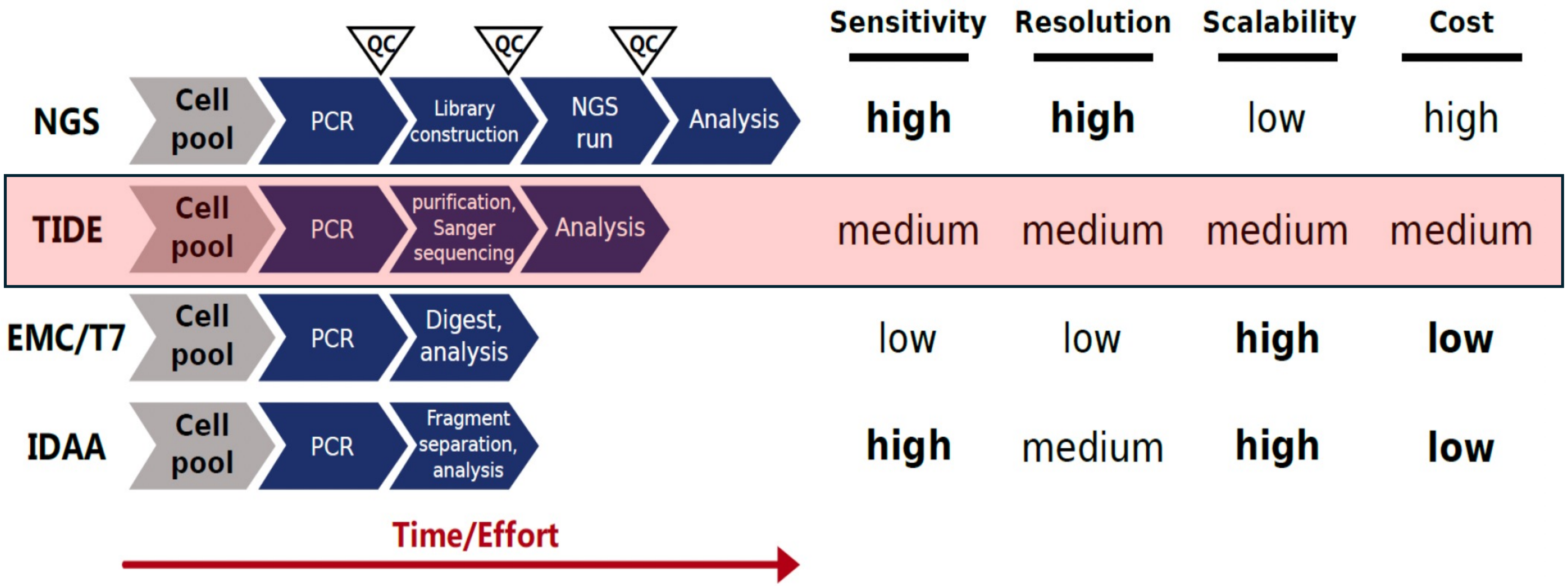
- The methodology is based on a novel tri-primer PCR amplification method and capillary electrophoretic fragment analysis using a standard DNA sequenator instrument.

Indel detection by amplicon analysis (IDAA)



Understanding the outcomes of genome editing

TIDE – Tracking of Indels by Decomposition



Understanding the outcomes of genome editing

TIDE – Tracking of Indels by Decomposition

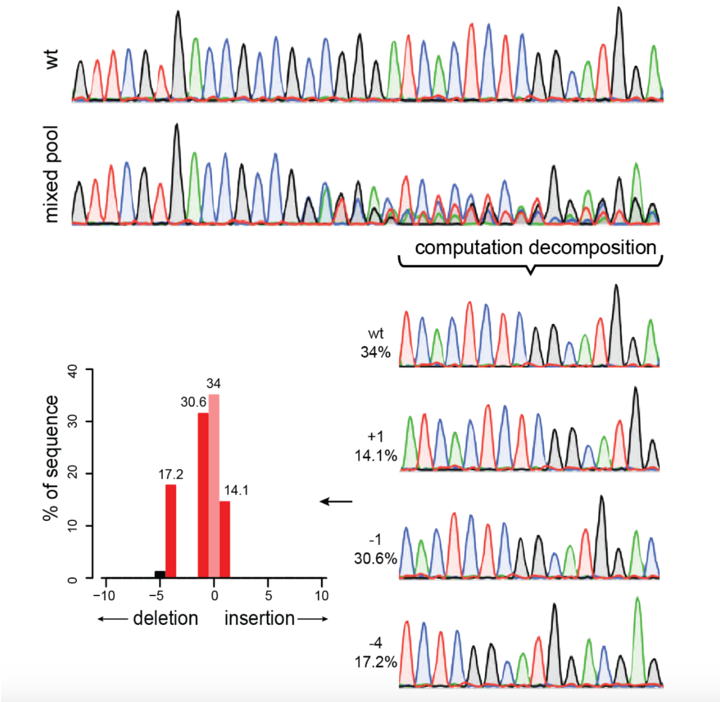
- TIDE is a simple and accurate assay to precisely determine the spectrum and frequency of targeted mutations generated in a pool of cells by genome editing tools such as CRISPR/Cas9, TALENs and ZFNs.

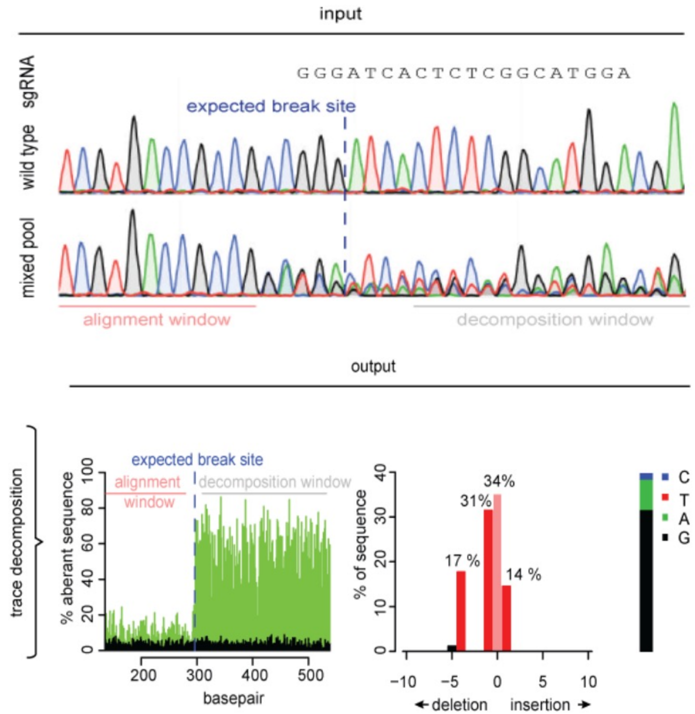
- TIDE requires only standard molecular biology reagents and involves three simple steps:
  - One pair of standard PCR reactions.
  - One pair of standard capillary ("Sanger") sequencing reactions.
  - Analysis of the two resulting raw sequencing files using the TIDE web tool.

Understanding the outcomes of genome editing

TIDE – Tracking of Indels by Decomposition

TIDE	Cell pool	PCR	purification, Sanger sequencing	Analysis	medium	medium	medium	medium





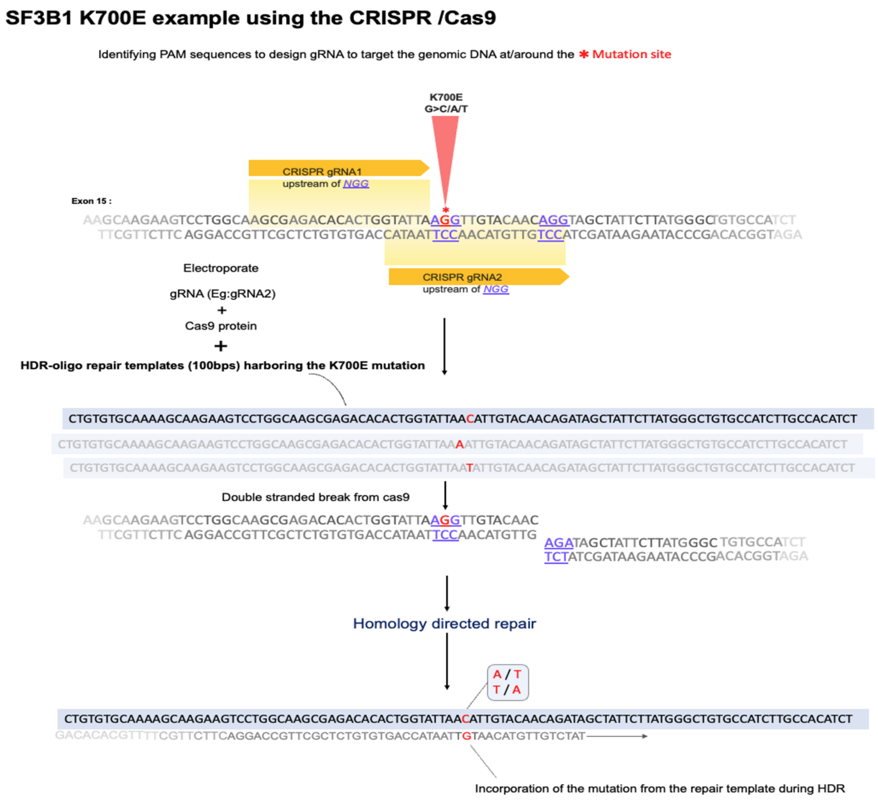
Understanding the outcomes of genome editing

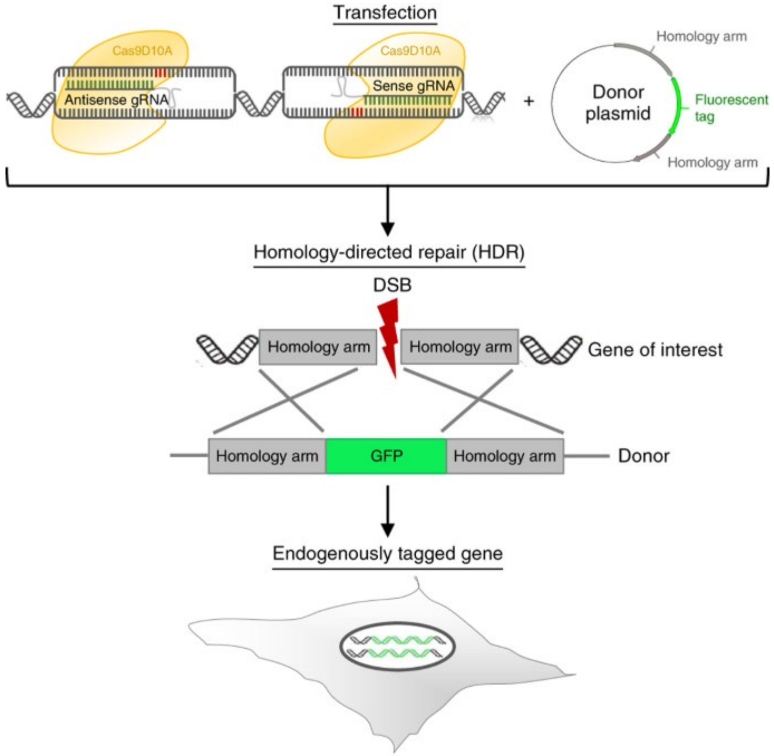
TIDE – Tracking of Indels by Decomposition

- **Let us analyze some of the data online**

Determining knock-in/ nucleotide variations

Gene knock-in thorough HDR (Homology Directed “Repair)

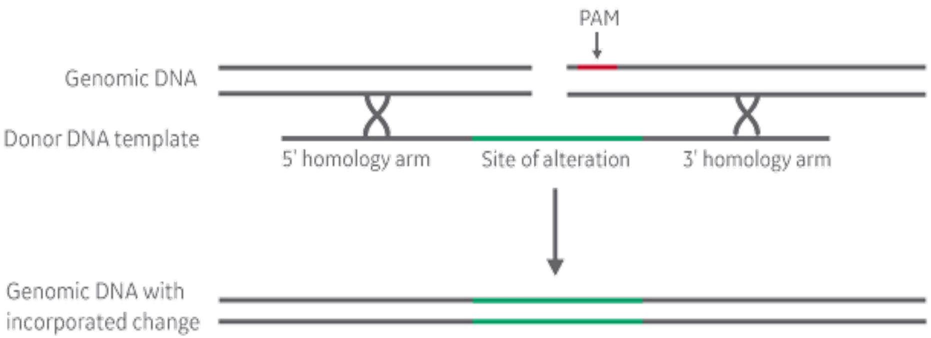


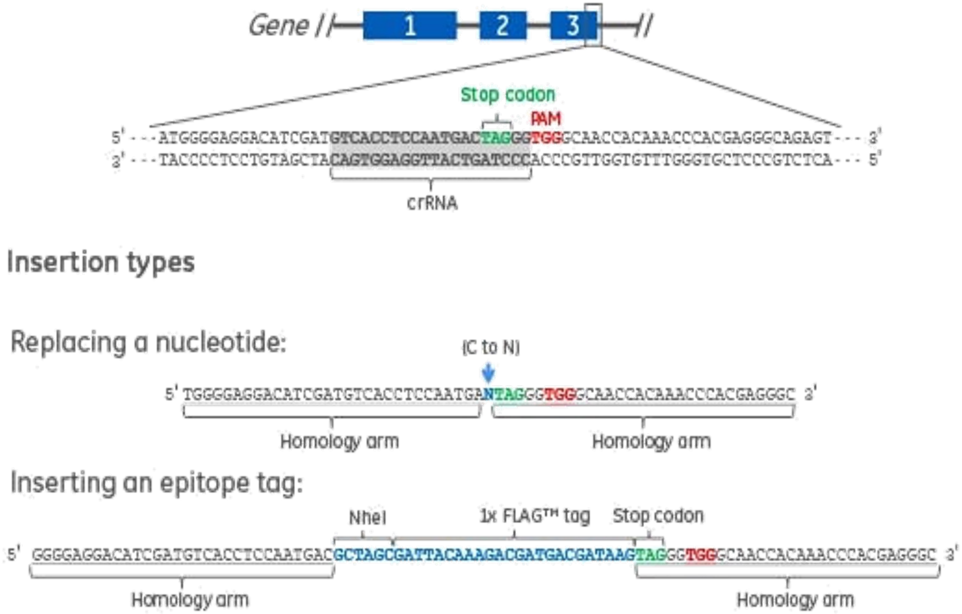


Determining knock-in/ nucleotide variations

SNP knock-in thorough HDR (Homology Directed Repair)

- Example : Creating a base change in a gene



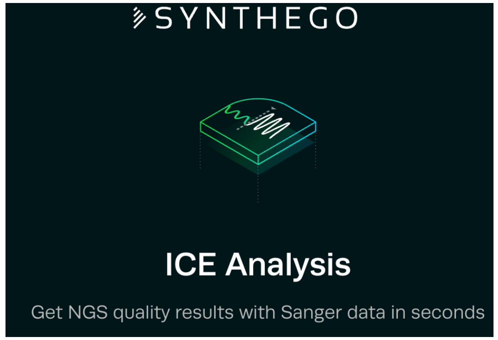


- We perform the editing in cells and pick single clones of these edited cells to scrutinize for knock-in
- Multiple rounds of preliminary QCs

Determining knock-in/ nucleotide variations

SNP knock-in thorough HDR (Homology Directed Repair)

- Example : Creating a base change in a gene
    - Editing and recombination efficiencies
    - - Assessed bulk edited population using
    - T7 endonuclease assay
    - TIDE /TIDER (Tracking of Indels – insertions/ deletions / recombinations)



T7 endonuclease assay

Lab based assay
PCR
T7 endonuclease enzyme
Run gel and assess editing efficiency

Indel tracking

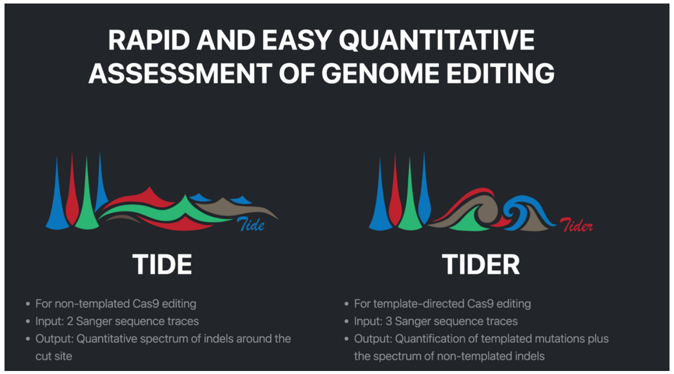
Sanger Sequencing based assay
PCR
Send for sanger sequencing
Analyze through input of sequence files

Synthego : Inference of CRISPR Edits

https://ice.editco.bio/#/

TIDE :

https://tide.nki.nl/



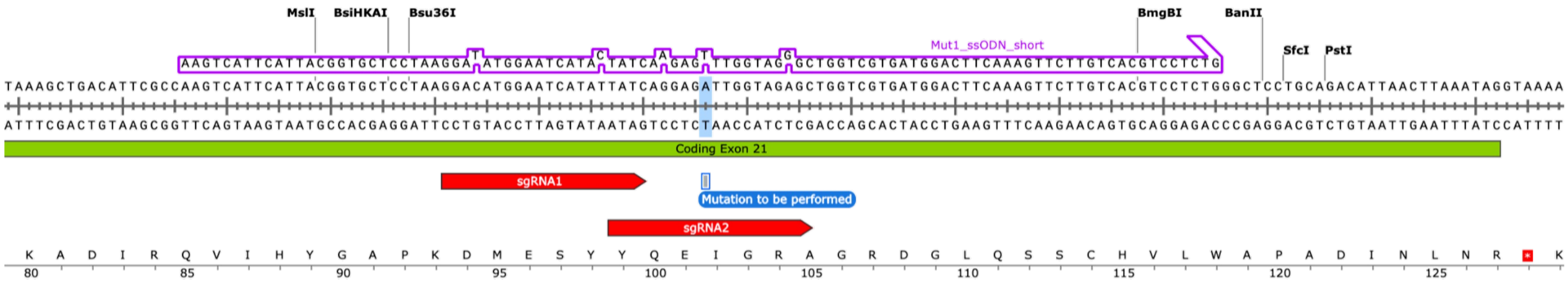
Determining knock-in/ nucleotide variations

SNP knock-in thorough HDR (Homology Directed Repair)

- Example : Creating a base change in a gene

Editing and recombination efficiencies

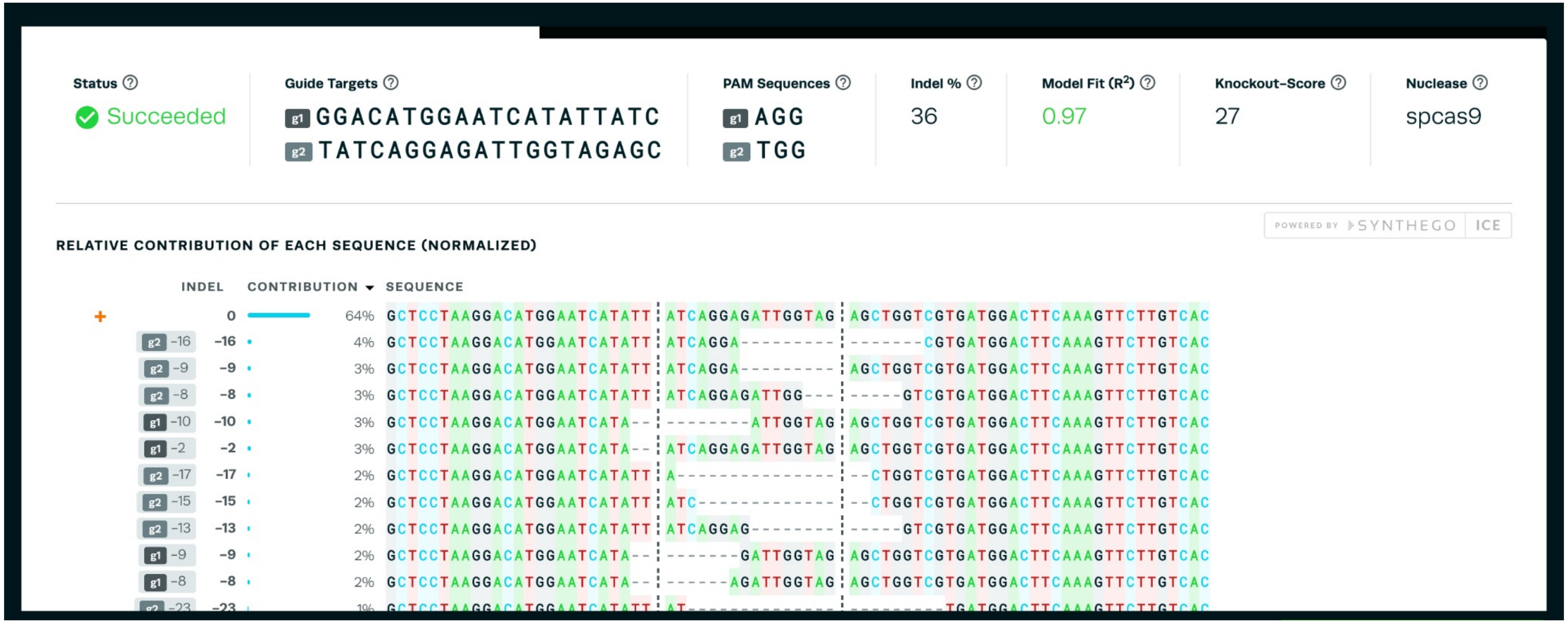
Exon 21



Determining knock-in/ nucleotide variations

SNP knock-in thorough HDR (Homology Directed Repair)

- Example : Creating a base change in a gene



Understanding the outcomes of genome editing

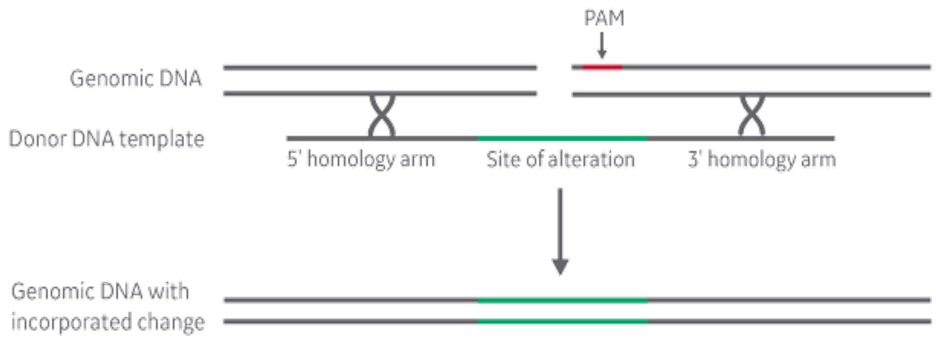
Recombination events – Inference of CRISPR edits

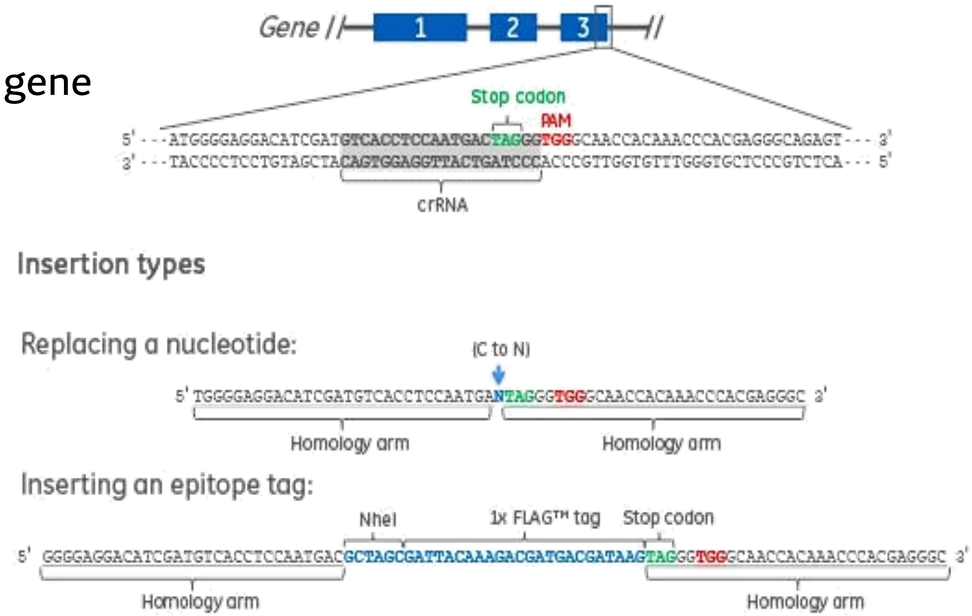
- **Let us analyze some of the data online**

Determining knock-in/ nucleotide variations

Gene knock-in thorough HDR (Homology Directed Repair)

- Example : Tagging in a fluorescent reporter to DLX5 gene





- We perform the editing in cells and pick single clones of these edited cells to scrutinize for knock-in
- Multiple rounds of preliminary QCs
- Process takes from anywhere from about 15 to 30 days to obtain an inserted/ knock-in line

Understanding the outcomes of genome editing

Tagging genes –

- **Let us analyze some of the data online**

In Summary

We looked into

- Understanding effects of double strand break created by Cas9
- How can we
  - Analyse them
  - Quantify them
  - Use information to choose efficient guides - depends on cell lines / cell state
  - Use information to determine success rate of desired outcome

- We also briefly discussed on
  - Understanding data coming out of sanger sequencing
  - Using sanger sequencing data to determine knock-in / SNP generation and gene-tagging
  - Various QC methods used at various stages of the process

- Q and A – quiz format

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questions?

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